EEE303 Tutorial 7

1. Referring to Figure T7-1, a capacitor-run motor has a NEMA rating of 30 milli-horsepower. If the capacitor-run motor operates at full-load, calculate the following:

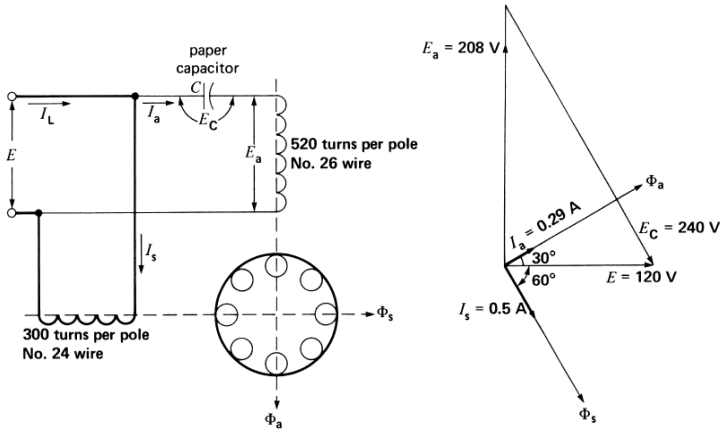


Figure T7-1

a. The line current I_L

b. The power factor of the motor

c. The active power absorbed by each winding

d. The efficiency of the motor

2. Referring to Figure T7-2, the effective impedance of the main and auxiliary windings under locked-rotor conditions are given as follows:

	Effective resistance	Effective reactance
Main winding	4 Ω	7.5 Ω
Auxiliary winding	7.5 Ω	4 Ω

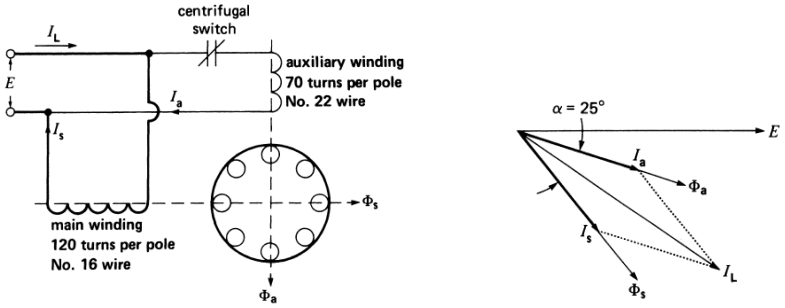


Figure T7-2

If the line voltage is 119 V, calculate the following:

a. The magnitude of I_a and I_s

b. The phase angle between I_a and I_s

c. The line current I_L

d. The power factor under locked-rotor conditions

3. Refer to Table A and calculate the following:

TABLE A CHARACTERISTICS OF A CAPACITOR-START MOTOR

Rating: 250 W, 1760 r/min, 115 V, 60 Hz, Insulation Class 105°C

	Full-load			No-load	
voltage	–	115 V	voltage	–	115 V
power	–	250 W	current	–	4.0 A
current	–	5.3 A	losses	–	105 W
P.F.	–	64%		Locked rotor	
efficiency	–	63.9%			
speed	–	1760 r/min	voltage	–	115 V
torque	–	1.35 N·m	current I_s	–	23 A
	Breakdown		current I_a	–	19 A
			current I_L	–	29 A
torque	–	3.4 N·m	torque	–	6 N·m
speed	–	1600 r/min	capacitor	–	320 μF
current	–	13 A			

a. The voltage across the capacitor under locked-rotor conditions

b. The corresponding phase angle between I_s and I_a

4. Refer to the 6 W, shaded-pole motor in Table B and calculate the following:

TABLE B

Properties of a Shaded-Pole Motor, Having 2 poles, Rated 6 W, 115 V, 60 Hz.

No-load		
current	0.26	A
input power	15	W
speed	3550	r/min
Locked rotor		
current	0.35	A
input power	24	W
torque	10	mN·m
Full-load		
current	0.33	A
input power	21	W
speed	2900	r/min
torque	19	mN·m
mechanical power	6	W
breakdown speed	2600	r/min
breakdown torque	21	mN·m

a. The full-load power factor

b. The slip at the breakdown torque

c. The per unit no-load current and locked rotor current

5. A 230 V, 50 Hz, 8-pole, single-phase induction motor has the following parameters;
$R_1 = 2\ \Omega$, $X_1 = 1\ \Omega$, $R_2 = 4\ \Omega$, $X_2 = 1.5\ \Omega$, $X_m = 180\ \Omega$.
The motor runs at 720 rpm on load. Calculate the output power, output torque and efficiency. The friction and windage loss including core loss is 55 Watts.

6. A stepper motor advances 2.5° per step. How many pulses are needed to complete 8 revolutions?

7. A permanent magnet stepper motor that advances 30° per step is driven by a series of pulses having a duration of 20 ms. How long will it take for the rotor to make one complete revolution?